

Year :

Diary

Subj : UNIDESK

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UniDesk - Mini Internal Ticketing System

Employee

↓
Login

↓
Create/View Ticket

↓
IT support sees ticket

↓
Ticket gets updated

↓
Comments are added

↓
Problem gets Resolved.

Data

User

Tickets

Comments

PostgreSQL Database

Fast API

React Frontend

JWT authentication

Ticket CRUD

Input validation

API test

70%+ API test coverage.

Year :

2020

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Goal : Working Ticketing System

- i) Project Goal
- ii) Problem statement
- iii) scope
- iv) Users and Roles
- v) Core Features
- vi) Authentication
- vii) Ticket Requirements
- viii) Comment Requirements
- ix) Database Requirements
- x) API Requirements.
- xi) Validation Rules
- xii) Autorization Rules
- xiii) Frontend Requirements
- xiv) Error Handling
- xv) Testing Requirements.
- xvi) Out of scope
- xvii) Defination of Done.

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Defination of Done:

1. User can authenticate
2. Authorized user can create a ticket
3. User can view appropriate tickets.
4. Ticket can be updated.
5. Ticket can be deleted according to permission.
6. Comments work.
7. Validation works.
8. Unauthorized requests are rejected
9. React Client work with API
10. API test pass.
11. Coverage is above 70%.

Alice Johnson → abcdEfgh2

0 → AbcdEfgh2

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1. Dashboard empty-state check

Dashboard view.

For the tiny task 1, and it is a one time UI task. so Agent md did not make the coding itself significantly better. The change I was simple enough either way.

Without agent, Earlier task also caused confusion about migration, database state, and ambiguous lifecycle rules.

with agent.

- i) Inspect the existing implementation before editing
- ii) Check the request behavior already exists
- iii) Make the smallest possible change.
- iv) Avoid changing API, routing or state logic
- v) Run focused validation
- vi) Report exactly which file changed.

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2. Add Dashboard Ticket Search.

Task 2 Feature was very similar to the
times.

- ① Dashboard ৭ ticket search শাকব।
- ② Search and status filter একসাথে কাজ কৰব।
- ③ Search কৰ কোন ticket পাবো না হোল
new empty message দেখাব।
- ④ status filter ৭ৰ কাৰনে কোন ticket না থাকিল
আগৰ empty message টা-ই শাকব।
- ⑤ দুই ধৰনেৰ empty state আলাদা হাৰাত হব।

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3. Enforce Ticket Lifecycle Transition.

Task-3 was noticeably easier to implement with Agent.md, mainly because it provided clear decision-making rules.

The final feature was not dramatically more complex with Agent.md. the code change was still small. The biggest improvement was it reduce uncertainty.

Without: The hardest part was deciding what the requirement meant.

with: The hardest part was mostly implementation and verification.

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2. Implement Ticket Assignment and Agent Work Queues.

A Fullstack Feature where

- ① Create new relationship with Ticket.
- ② Create new migration for database.
- ③ Create protected endpoint so that only authorized user/agent can access it.
- ④ Adding role-based permission. Who can assign ticket who can see that etc.
- ⑤ Adding necessary filter
- ⑥ Adding necessary control in Dashboard UI for assigning ticket.

Agent যেন ticket নিজের কাছে assign করতে পারে
and নিজের assign করা ticket সূন্যের আলাদা
work queue দেখতে পারে। এমন frontend + backend
feature create করা

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5. Harden Authentication and session Management

i) JWT validation strong করা - token যাচাই
ও valided কি না check করা -

ii) Centralized role dependencies create করা.
User এর role অনুযায়ী permission check
করার একটি কেন্দ্রীয় ব্যবস্থা।

iii) Startup Configuration validation add করা.
Application start হওয়ার সময় important
configuration/security settings যাচাই লাগে
কি না check করা।

iv) Frontend session handling Improve করা.
Frontend এ login session/token perfectly
manage করা।

Full application এর authentication and login session
এর security বাড়ানো করা, যাতে unauthorized
user access করতে না পারে and session-related
না করা।

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Task-3

With agent.md thoes business policy is already been declared in the Agent.md file. so the task become implementation and verification rather than policy invention.

Task-4

In one sentence: On this codebase, the context file did not make the agent smarter it made the agent's unsupervised guesses fewer and on the two task large enough for guesses to compound, that difference was the entire story.

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i) Umidesk is a full-stack internal IT support ticketing system.

How ii) Frontend is react which communicate with a FastAPI backend.

iii) The backend uses SQL Alchemy to interact with PostgreSQL.

iv) Pydantic handles request validation.

v) JWT handles authentication.

vi) Role-based authorization controls employee and support agent action.

vii) Alembic manages database migration, and pytest is used for backend testing.

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Backend API level to must attack test

1. Missing Input Attack
2. Empty Input
3. Null Input
4. Whitespace Input
5. Invalid data type
6. Boundary value
7. Very Large Input
8. Special character
9. Duplicate Data
10. Invalid Format
11. Invalid Id
12. Non-existing Resource
13. Unauthorized Access
14. Authentication Bypass
15. Authorization / RBAC
16. Token Manipulation
17. Expired Token
18. Wrong credentials
19. SQL Injection
20. XSS Attack
21. Parameter Tampering
22. HTTP Method Manipulation
23. Rate Limit / Brute Force Attack
24. Concurrent Request Attack
25. Error Handling

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26. Unexpected API Response
27. File upload
28. Session Handling
29. Data Leakage
30. CORS/Security Header Attack.

Not biendly, I would verify the result agent's ~~test output~~ logs, traceback and the actual system behavior.